Patient Waiting Time Analysis and Optimization in Hospital Using Business Analytics

A Research Project on Enhancing Healthcare Efficiency through Data-Driven Insights

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Introduction

The Challenges

- Patient waiting time is a critical indicator of healthcare quality and patient satisfaction.
- Prolonged delays lead to patient frustration and can negatively impact clinical outcomes.
- Inefficient patient flow often results in resource underutilization and increased operational costs.

Business Analytics Role

- Leveraging data mining and predictive modeling to identify hidden bottlenecks in the system.
- Using statistical analysis to optimize staffing levels and scheduling based on patient arrival patterns.
- Transforming raw hospital data into actionable insights for operational excellence.

Hospital Profile

Dr. Karad Hospital is a healthcare institution located in Satpur, Nashik, providing medical services to patients from nearby urban and rural areas. The hospital offers various healthcare facilities including outpatient consultation, registration, diagnostic services, treatment, and billing services.

The hospital focuses on delivering quality healthcare services with proper patient care and operational efficiency. Due to increasing patient flow and service demand, managing patient waiting time and maintaining smooth hospital operations have become important aspects of hospital management.

The hospital uses both manual and digital processes for patient management and service delivery. The study was conducted in this hospital to analyze patient waiting time, identify operational inefficiencies, and suggest optimization strategies using business analytics techniques.

Objectives of the Study

- To analyze the waiting time at different stages of the hospital service process, including registration, consultation, diagnostic services, pharmacy, and billing, in order to identify the major sources of delay
- To identify the operational factors responsible for increased patient waiting time, such as patient flow, staff availability, resource utilization, appointment scheduling, and process inefficiencies.
- To apply business analytics techniques for evaluating hospital operations and recommend data-driven strategies to optimize workflows, reduce waiting time, and improve operational efficiency.
- To assess patient satisfaction regarding hospital waiting time and service quality, and examine how waiting time influences the overall patient experience and healthcare delivery

Literature Review

Current Trends

Studies indicate that over 30% of patient dissatisfaction stems from long waiting periods, highlighting wait time as a primary quality metric in modern healthcare.

Technological Integration

The integration of Business Intelligence (BI) tools and AI enables real-time patient tracking and predictive resource allocation to mitigate delays.

Analytical Frameworks

Previous research emphasizes the efficacy of Queuing Theory and Lean Six Sigma methodologies in streamlining hospital workflows and reducing non-value-added time.

Gap Analysis

There is a significant need for localized studies in Indian management contexts to address specific infrastructural challenges and cultural arrival patterns.

Research Methodology

Research Design

A combination of **Descriptive** and **Analytical** research designs to understand current patterns and identify optimization opportunities.

Data Collection

- **Primary:** Observation,Patient Feedback,Staff Intereaction,Google Form.
- **Secondary:** Research journals,WebsitesPrevious research papers on waiting time analysis and healthcare analytics.

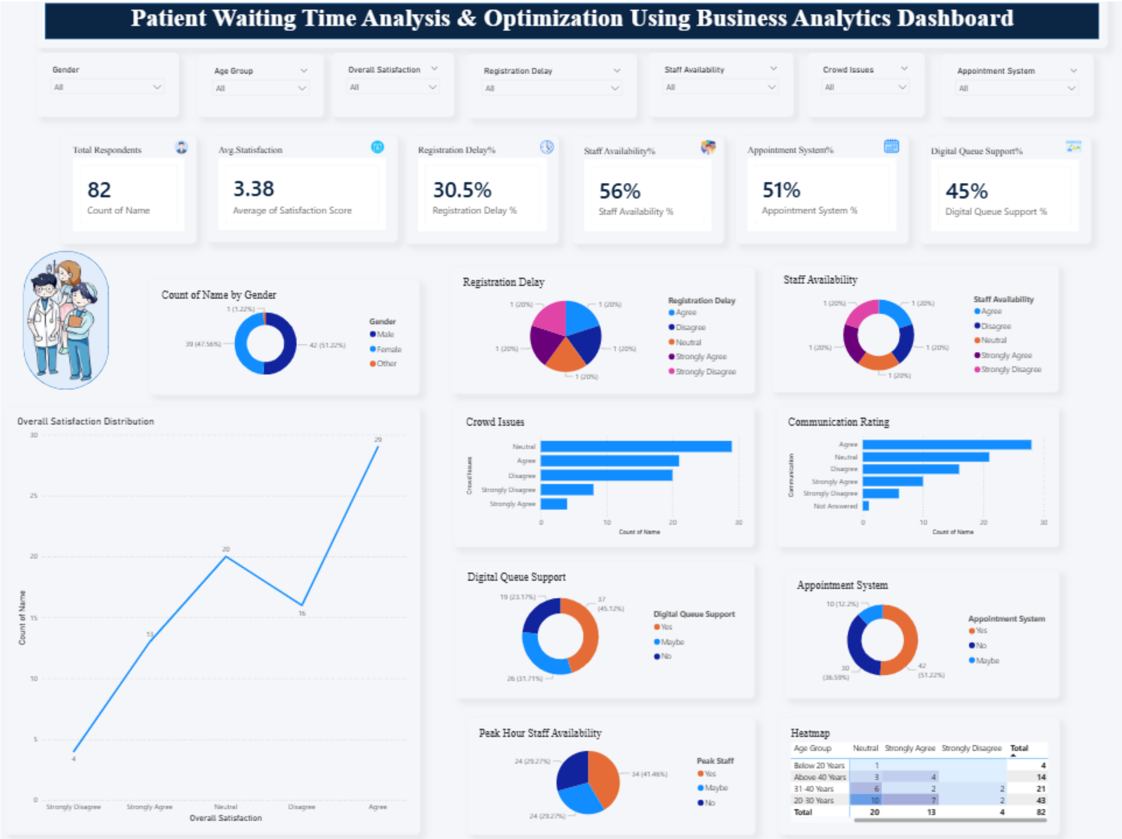
Analytical Tools

- **Statistical:** Pivot Tables,Comparative Analysis
- **Visualization:** Excel,Bar Charts,Pie Charts,Column Charts, PowerBi,Donut Charts.

Sampling Technique

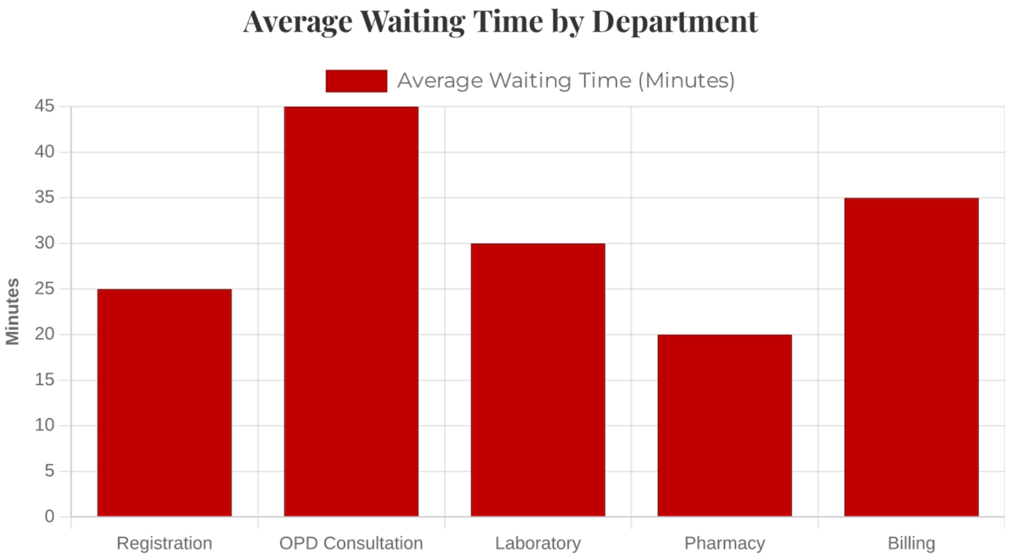
Convenience sampling technique was used for the study. The respondents were selected based on their availability and willingness to participate during the hospital visit.

Data Analysis and Interpretation



- The dashboard was created using Power BI, Pivot Tables, Pivot Charts, and Slicers to analyze patient waiting time and hospital operational efficiency dynamically.
- The dashboard provides visual insights regarding patient demographics, waiting time satisfaction, crowd management issues, digital queue system support, registration delays, appointment system effectiveness, and staff availability impact.
- Interactive slicers were used to filter data dynamically based on age group, gender, and patient responses. The dashboard helps in identifying operational bottlenecks and supports data-driven decision-making for improving hospital service efficiency.

Data Analysis and Interpretation



Waiting Time Distribution

Significant variance observed across departments, with OPD and Billing showing the highest delays.

Correlation Analysis

A positive relationship is observed between patient arrival volume and total processing time. As patient arrivals increase, processing time also tends to increase.

Peak Hour Identification

Data identifies 10:00 AM - 1:00 PM as the critical peak period requiring maximum resource allocation.

Findings / Results

Registration Bottlenecks

Registration and billing processes are the primary bottlenecks, accounting for approximately 40% of total patient wait time.

Staffing Mismatch

Peak patient arrival times often do not align with maximum staff availability, leading to significant service delays during high-demand periods.

Process Redundancy

Duplicate data entry points across different departments contribute to unnecessary administrative delays and increased error rates.

Satisfaction Impact

A direct inverse correlation exists between wait time and patient satisfaction scores, with a sharp decline after the 30-minute mark.

Conclusion

"Business analytics provides a powerful lens to transform hospital operations from reactive to proactive, ensuring that patient care remains the central priority."

Operational Summary

Optimization through data-driven insights significantly reduces wait times and enhances patient flow across all departments.

Quality Impact

Reduced waiting times directly correlate with improved clinical outcomes and higher patient satisfaction scores.

Final Thought

Data-driven decision-making is no longer optional but essential for the sustainability of modern healthcare management.

Optimization Suggestions

Predictive Scheduling

Utilize historical data to forecast patient arrival patterns and align staffing levels dynamically with peak demand periods.

Digital Transformation

Implement self-service kiosks and mobile registration platforms to decentralize the intake process and reduce front-desk congestion.

Process Re-engineering

Adopt Lean Six Sigma principles to identify and eliminate non-value-added steps, streamlining the end-to-end patient journey.

Real-time Monitoring

Deploy interactive BI dashboards for administrators to monitor live patient flow and intervene immediately when bottlenecks occur.

Learning from Project

Analytical Skills

Gained hands-on proficiency in data collection, cleaning, and advanced visualization techniques using Excel and BI tools.

Domain Knowledge

Deepened understanding of healthcare operations, patient flow dynamics, and hospital experience management.

Problem Solving

Learned to apply theoretical business analytics concepts to solve complex, real-world operational challenges in a hospital setting.

Professional Growth

Developed critical insights into the strategic role of a Business Analyst in driving efficiency within the healthcare sector.

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THANK YOU

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